

TMDB Movie Data Analysis using Pandas and APIs Lab

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1. Methodology and Data Preprocessing

The project implemented a complete workflow using API data extraction from The Movie Database (TMDB). The primary objective was to evaluate movie performance through financial, audience, and production-related metrics.

- **Data Cleaning & Transformation:** The raw dataset was processed by removing irrelevant columns and transforming nested JSON-like fields (such as genres, production companies, and spoken languages) into a structured format.
- **Feature Engineering:** Technical columns were created for Profit (Revenue - Budget) and ROI (Revenue / Budget). Financial values were normalized into millions of US dollars (MUSD) for analytical consistency.
- **KPI Implementation:** A user-defined function (UDF) was developed to rank movies across multiple performance indicators, including ROI, vote counts, and popularity scores.

2. Key Performance Indicators (KPIs) and Results

The analysis identified Avatar as the premier financial performer, holding the records for Highest Revenue.

The data demonstrated that high production expenditure does not strictly correlate with high financial efficiency (ROI). Audience engagement was highest

for The Avengers (Most Voted/Popular), while Avengers: Infinity War achieved the

Highest Rated status.

3. Advanced Filtering and Comparative Analysis

- **Search Queries:** Advanced filtering for specific actor-director pairings (Bruce Willis and Quentin Tarantino) yielded no results, highlighting a limitation caused by the predefined sample size of 18 movies.

- **Franchise vs. Standalone:** Utilizing aggregation, the study found that standalone movies outperformed franchise movies in Mean Revenue and Median

ROI. Franchises exhibited higher Mean Popularity and Mean

Budget, suggesting that brand recognition drives audience awareness but does not inherently guarantee superior financial margins.

4. Franchise and Director Success

The Avengers Collection was the most successful aggregation by total revenue.

In director-based metrics, James Cameron generated the highest total revenue, while Anthony Russo maintained the highest mean rating, proving that

commercial success and critical evaluation can vary significantly by creator.

5. Visualization and Conclusion

The project utilized scatter plots to investigate the relationship between Revenue and Budget, and popularity-versus-rating plots to compare audience

attention with qualitative evaluation. ROI distributions were also analyzed across genres.

Conclusion: The workflow successfully demonstrated how structured API data can

be transformed into actionable insights through feature engineering and KPI calculation. Despite limitations such as a small sample size, the project provides a robust framework for investigating the financial and critical dynamics of the global movie industry.